Delta-Scale 10-m Tidal Flat Topography Reconstruction: A Transferable Deep Learning Approach Using Sentinel-2 Time Series

Peng Li, Jiahan Zhang, Fengling Yu, Kai Tan, Nan Xu, Chunpeng Chen, Pengfei Tang, Cunren Liang, *Member, IEEE*, Zhenhong Li, *Senior Member, IEEE*, and Houjie Wang

***Abstract*—Reliable tidal flat elevation data with a resolution better than 30 m are vital for understanding macroscale sedimentation processes and storm surge erosion in estuarine deltas. However, the highly dynamic and turbid environments of global estuaries pose significant challenges for traditional manual or uncrewed aerial vehicle (UAV) methods in delta-scale monitoring. Insufficient elevation data currently hinder sea-level rise modeling and coastal management for the world's densely populated, flood-prone river deltas. There is an urgent need to characterize and reconstruct the vertical geomorphology of tidal flats through improved observations and modeling. In this study, we propose a robust tidal flat topography mapping framework using freely available satellite imagery and deep learning (DL). First, we developed an enhanced U-Net model to delineate land–water boundaries and generate inundation frequency maps from time-series Sentinel-2 imagery. A regionally adaptive nonlinear inversion model was then constructed using inundation frequency and ICESat-2 elevation data, allowing us to generate a 10-m resolution tidal flat digital elevation model (DEM) of the Yellow River estuary. The DL model achieved an $F1$-score of 0.96, and the resulting DEM showed strong agreement with UAV LiDAR data, yielding a root mean square error (RMSE) of 0.18 m. The reconstructed elevations range from −0.51 to 0.96 m, exhibiting a spatial gradient of lower elevations in the north and higher elevations in the southern and eastern sectors. To assess transferability, we applied the method to the radial sand ridge system in the South Yellow Sea. The model achieved a comparable RMSE of 0.52 m without retraining, demonstrating robust generalization across different estuarine environments. This study provides a cost-effective framework for monitoring coastal geomorphological changes and helps fill the data gap in delta-scale topographic mapping for global intertidal zones.**

***Index Terms*—Coastal topography, deep learning (DL), digital elevation model (DEM), river delta, satellite remote sensing.**

Received 5 September 2025; revised 9 February 2026 and 6 April 2026; accepted 18 April 2026. Date of publication 21 April 2026; date of current version 5 May 2026. This work was supported in part by Shandong Provincial Natural Science Foundation under Grant ZR2024MD061 and in part by the Natural Science Foundation of China under Grant 42041005-4 and Grant 42571378. *(Corresponding author: Peng Li.)*

Peng Li, Jiahan Zhang, and Houjie Wang are with the Institute of Estuarine and Coastal Zone, College of Marine Geosciences, Key Laboratory of Submarine Geosciences and Prospecting Technology, Ministry of Education, Ocean University of China, Qingdao 266100, China, and also with the Laboratory for Marine Geology, Qingdao Marine Science and Technology Center, Qingdao 266237, China (e-mail: pengli@ouc.edu.cn; zhangjiahan@stu.ouc.edu.cn; hjwang@ouc.edu.cn).

Fengling Yu is with the College of Ocean and Earth Sciences, State Key Laboratory of Marine Environmental Science, Fujian Provincial Key Laboratory for Coastal Ecology and Environmental Studies, Dongshan Swire Marine Station, Xiamen University, Xiamen 361102, China (e-mail: yfhu@chd.edu.cn).

Kai Tan is with the State Key Laboratory of Estuarine and Coastal Research, East China Normal University, Shanghai 200241, China (e-mail: ktan@sklec.ecnu.edu.cn).

Nan Xu is with the Key Laboratory for Geo-Environmental Monitoring of Great Bay Area, Ministry of Natural Resources and Guangdong Key Laboratory of Urban Informatics, Shenzhen University, Shenzhen 518060, China (e-mail: xunan2025@szu.edu.cn).

Chunpeng Chen is with the Department of Land Surveying and Geoinformatics, Hong Kong Polytechnic University, Hong Kong, China, and also with the State Key Laboratory of Estuarine and Coastal Research, East China Normal University, Shanghai 200241, China (e-mail: chunpeng.chen@polyu.edu.hk).

Pengfei Tang is with the Research Institute of Agriculture and Life Sciences, Seoul National University, Seoul 08826, Republic of Korea (e-mail: pengfei@snu.ac.kr).

Cunren Liang is with the Institute of Remote Sensing and Geographic Information System, School of Earth and Space Sciences, Peking University, Beijing 100871, China (e-mail: cunren.liang@pku.edu.cn).

Zhenhong Li is with the College of Geological Engineering and Geomatics, Chang'an University, Xi'an 710054, China (e-mail: zhenhong.Li@chd.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3686116

I. Introduction

TIDAL flats, generally defined as unvegetated mud or sand flats situated between mean high and low tide levels, are distinct from vegetated intertidal systems like mangroves and salt marshes [1], [2]. Characterized by periodic tidal inundation, these landforms support diverse biological communities and facilitate key coastal processes [3], [4], including nutrient cycling, carbon sequestration, and providing breeding grounds for marine species [5], [6]. They also serve as natural buffers that mitigate storm surges and coastal erosion, protecting coastal infrastructure [7]. However, climate change and anthropogenic pressures such as sea-level rise, coastal development, and pollution contribute to the degradation of these ecosystems [8], [9], [10], [11]. Because tidal flat morphology and sediment dynamics respond to shifts in sea level, sediment supply, and hydrodynamic conditions, they serve as indicators of environmental change [12]. Consequently, mapping tidal flat topography is fundamental to quantifying the geomorphological and ecological evolution of these systems.

River deltas transport large sediment volumes to the coast, forming expansive tidal flats. Globally, 89 large deltas (> 1000 km^2) account for 84.3% of all deltaic land and 89% of the total deltaic population [13]. While over 11 000 deltaic features exist, including 885 medium (10–1000 km^2) and 1460 small (1–10 km^2) systems, these 89 deltas are the most geographically and socioeconomically significant [14]. Deltas

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:31:11 UTC from IEEE Xplore. Restrictions apply.

are typically low-lying, with over 100 000 km^2 situated below 2 m above mean sea level [15]. Currently, sediment starvation, land subsidence, and coastal flooding pose documented risks to these regions [11], [16], [17], [18], [19], [20]. However, existing global digital elevation models (DEMs) often lack the spatial resolution, coverage, and update frequency required for such dynamic environments [21]. The scarcity of high-resolution, delta-scale elevation data limits the capacity to model relative sea-level rise and affects the reliability of coastal management strategies [22], [23].

Various methods have been employed to map tidal flats, including traditional field surveys, remote sensing, and numerical modeling. Traditional techniques, such as real-time kinematic (RTK) positioning, provide centimeter-level vertical accuracy but are labor-intensive, which generally restricts their application to small study areas [24].

Synthetic aperture radar (SAR) offers the advantage of operating independently of solar illumination and weather conditions [25]. However, inherent speckle noise in SAR data can limit its efficacy compared with optical imagery for classifying complex coastal features [10], [26], [27], [28], [29]. For instance, multilooking Sentinel-1 Ground Range Detected (GRD) data to reduce speckle results in a 23-m spatial resolution, with shoreline uncertainties reaching approximately 46 m [30]. Additionally, high soil moisture and salinity in intertidal zones impede microwave penetration, while tidal dynamics increase temporal decoherence, complicating phase unwrapping and elevation extraction via traditional InSAR. More recently, the Surface Water and Ocean Topography (SWOT) mission. More recently, the SWOT mission has demonstrated potential for wide-swath water level measurements. However, its high-resolution raster product (L2_HR_Raster) currently uses a 100-m grid, which may not resolve fine-scale intertidal features [31], [32].

Uncrewed aerial vehicles (UAVs) and photogrammetric techniques provide high-resolution data but face significant operational constraints in coastal environments, including high winds, limited battery endurance, and the requirement to align flights with narrow low-tide windows [28], [33]. Consequently, satellite optical imagery, such as that from Sentinel-2, remains a primary tool for delta-scale mapping due to its systematic revisit times and multispectral capabilities.

A widely adopted approach using optical imagery is the waterline method, which reconstructs topography by extracting the land–water boundary across various tidal stages and correlating these lines with tide gauge data or hydrodynamic models [34], [35]. This technique has been applied to map tidal flats in diverse regions. Despite its utility, the method's performance depends on the precise identification of the waterline. Traditional hard-thresholding techniques using water or vegetation indices can lead to misclassification, particularly in highly turbid waters [36]. Furthermore, frequent cloud cover and sensor revisit cycles can limit the availability of cloud-free images across the full tidal range, potentially leading to data gaps in the resulting topographic products.

To address these limitations, researchers have applied machine learning (ML) and deep learning (DL) techniques [37], [38], [39], [40]. However, as optical imagery provides 2-D planar data, an external vertical reference is required to assign elevation values to extracted waterlines. While previous studies have utilized tide gauge records or hydrodynamic models [41], gauges are often spatially sparse in remote regions, and models can exhibit uncertainties in complex nearshore shallow waters due to bathymetric constraints [42]. Consequently, the vertical accuracy of tidal flat DEMs derived from these traditional approaches typically ranges from the decimeter to meter scale. While recent DL-based automation has improved waterline extraction, reported vertical root mean square errors (RMSEs) generally remain between 0.3 and 0.5 m, indicating potential for further optimization in large-scale monitoring. Additionally, model generalization and computational efficiency across diverse regions remain ongoing challenges.

Ground and airborne LiDAR systems provide high-resolution terrain elevation data but are constrained by limited spatial coverage and high operational costs at the delta scale [37]. Spaceborne LiDAR, such as the Advanced Topographic Laser Altimeter System (ATLAS) on ICESat-2, offers broader coverage by measuring surface elevation through laser pulse returns [43], [44]. This technology has demonstrated potential for tidal flat mapping [39], [45], [46], [47]. However, waterlines from single satellite images often fail to represent true contour lines due to spatial inconsistencies in water levels. To mitigate this, studies have employed inundation frequency (or water occurrence) maps derived from long-term Landsat or Sentinel-2 observations [33], [45], [48], [49]. These maps establish empirical relationships between inundation frequency and surface elevation. Nevertheless, challenges persist in estuarine environments with irregular tidal regimes and high turbidity, such as the Yellow River estuary, where the land–water boundary is often indistinct and tidal levels exhibit strong spatial heterogeneity [50], [51].

To address the aforementioned limitations, we propose a method to map delta-scale tidal flat topography at a 10-m spatial resolution by integrating time-series Sentinel-2 observations. We utilize a DL-based approach to generate inundation frequency maps and establish inversion models using ICESat-2 observations to derive the tidal flat DEM. Finally, the results are validated against airborne LiDAR data using nonparametric robust statistics, and the transferability of the model is evaluated across different intertidal environments.

II. Datasets and Methods

A. Study Area

The study area encompasses the core region of the Yellow River Delta National Nature Reserve, exceeding 500 km^2 (Fig. 1). Located on the eastern coast of China along the Bohai Sea, the Yellow River Delta is a dynamic coastal system shaped by the sediment load of the Yellow River [29]. The region is characterized by extensive tidal flats where the interplay of fluvial processes and tidal fluctuations creates a complex landscape. High water turbidity in this area often obscures the spectral distinction between land and water, while the combination of river discharge, tidal forces, and wave dynamics results in spatially variable tidal conditions [52].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:31:11 UTC from IEEE Xplore. Restrictions apply.

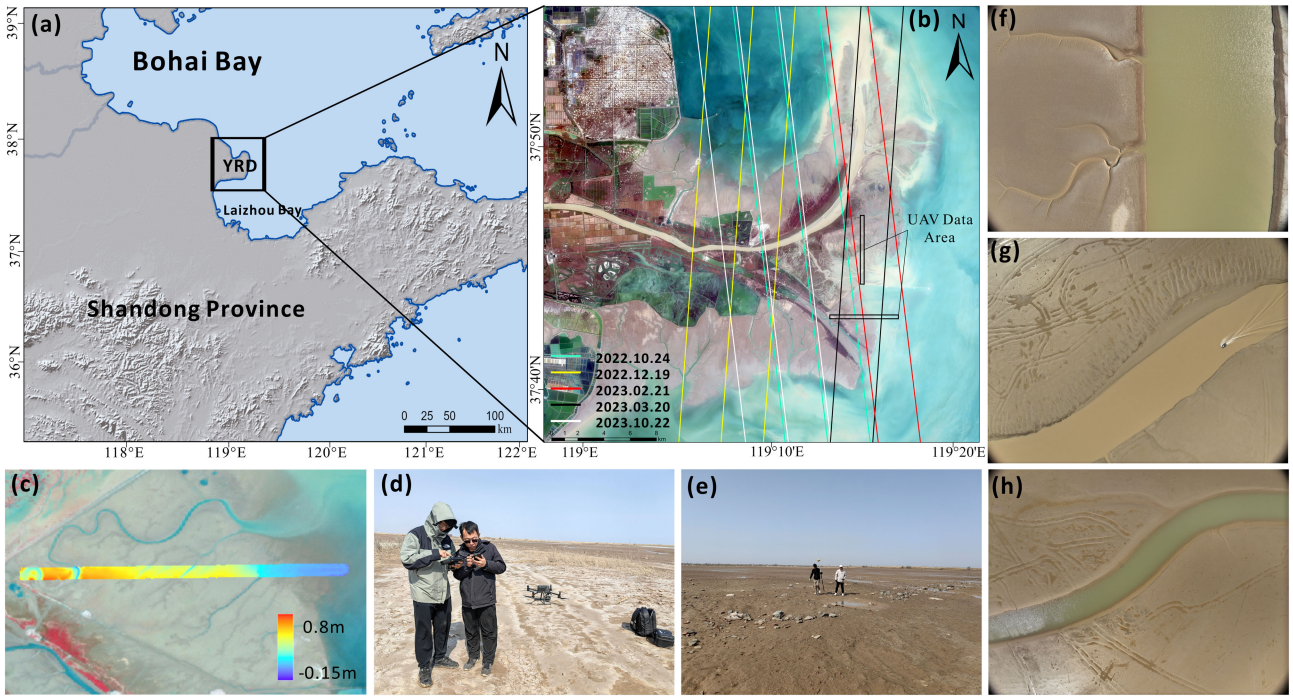


Fig. 1. (a) Study area location. (b) ICESat-2 laser tracks. Solid colored lines indicate observations from five dates in 2022 and 2023. Black rectangles denote UAV LiDAR coverage. (c) UAV LiDAR DEM. (d) and (e) UAV survey photographs. (f)–(h) UAV multispectral imagery.

TABLE I
SPECIFICATIONS OF DATASETS IN THIS STUDY

Datasets	Spatial Resolution	Vertical Accuracy	Temporal Coverage	Spatial Range
Sentinel-2	10 m	-	2022.01-2023.12	37.57°-37.93°N, 118.98°-119.34°E
ICESat-2	0.7 m	0.1 m	2022.01-2023.12	
UAV LiDAR	0.01 m	0.015 m	2024.03	
Tidal DEM	10 m	0.18 m	2022.01-2023.12	

B. Datasets

The Sentinel-2 mission consists of two satellites providing a combined revisit cycle of three to five days. The multispectral instrument features 13 bands with spatial resolutions up to 10 m (Table I). For this study, 221 Sentinel-2 Level 2A images acquired between January 1, 2022, and December 31, 2023, were processed via Google Earth Engine (https://developers.google.com/earth-engine/datasets/catalog/sentinel-2). Images with over 60% cloud cover were excluded, and cloud masking was performed using the QA60 band [53]. Analysis utilized bands B2, B3, B4, B8, and B11. The 20-m resolution B11 band was resampled to 10 m using nearest-neighbor interpolation to match the other bands.

ICESat-2 (https://nsidc.org/data/icesat-2) carries the ATLAS, which measures surface elevation using a multibeam LiDAR configuration. The system utilizes three pairs of beams, each consisting of a strong and a weak beam separated by 90 m, with 3 km spacing between pairs [54]. Due to the 4:1 energy distribution, strong beams provide a higher signal-to-noise ratio (SNR) in the optically complex conditions of tidal flats, whereas weak beam signals are often attenuated by moisture and shallow water. To minimize elevation uncertainty, only strong beam data from the ATL03 product were utilized. Based on local tidal records, 13 laser tracks acquired during low-tide windows between January 2022 and December 2023 were selected (Fig. 1). These tracks were partitioned into a modeling set (308 175 points) and a validation set (16 879 points).

UAV LiDAR data were employed for topographic validation. In March 2024, surveys were conducted during low tide using a DJI M300 RTK equipped with a Zenmuse L2 sensor. The survey areas, denoted by black rectangles in Fig. 1, cover strips approximately 200 m wide and 1000 m long, with a manufacturer-specified vertical accuracy of 1.5 cm +1 ppm.

C. Methods

The proposed delta-scale tidal flat mapping workflow consists of three modules (Fig. 2). Following data preprocessing, selected spectral and index features are integrated into a DL model for sea–land segmentation. Subsequently, waterlines are extracted to produce an inundation frequency map. Finally, by integrating elevation measurements from filtered ICESat-2 data, a nonlinear inversion model is constructed to generate the 10-m spatial resolution tidal flat DEM.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:31:11 UTC from IEEE Xplore. Restrictions apply.

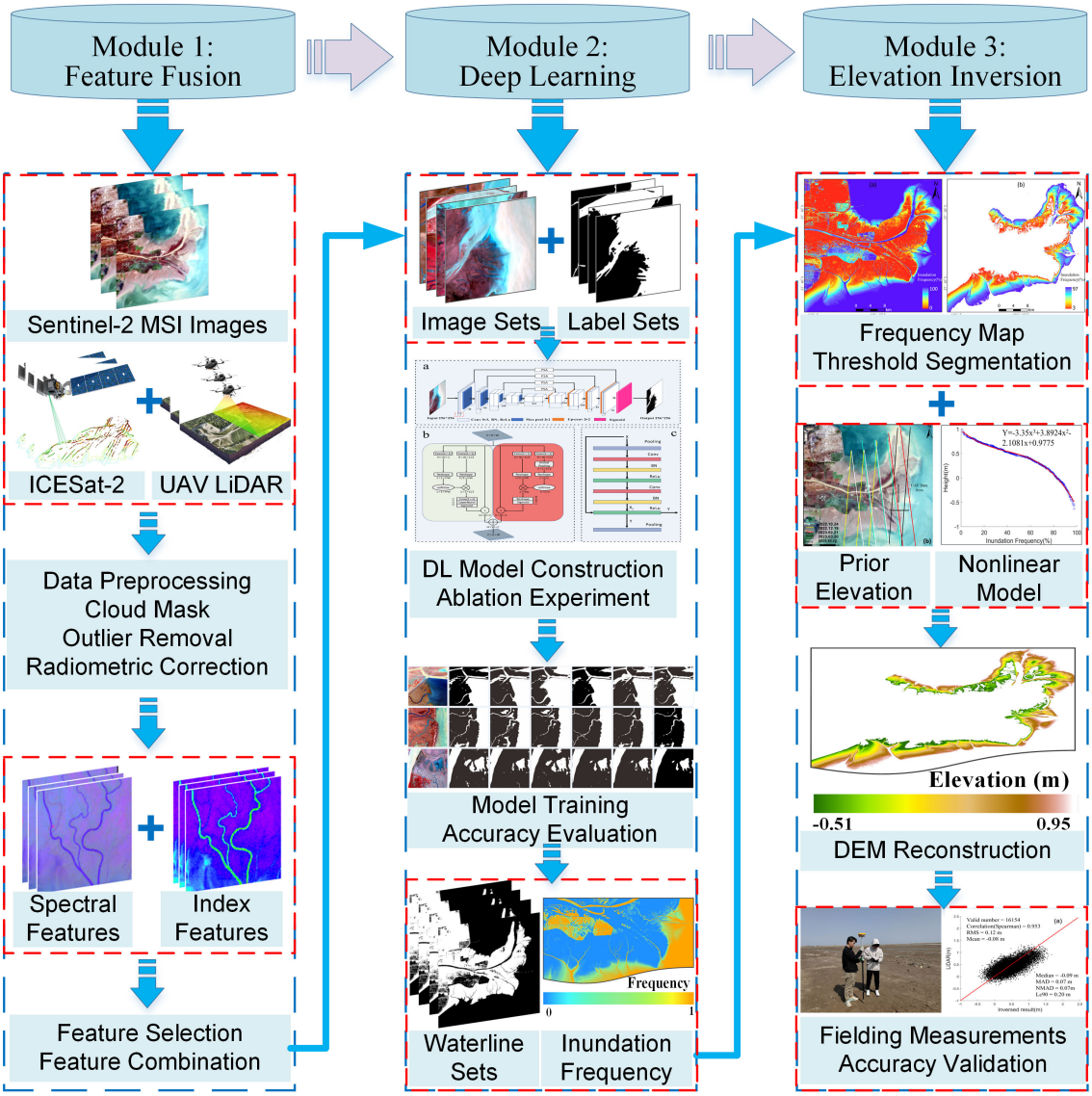


Fig. 2. Flowchart of the proposed tidal flat DEM mapping method.

1) DL Model Construction: Compared to single-index thresholding, DL models can improve classification performance by learning features across diverse datasets [55]. For this study, 24 Sentinel-2 images (one per month from January 2022 to December 2023) were selected to account for seasonal variations in water turbidity, sediment supply, and hydrodynamic conditions.

Standard metrics for identifying vegetation and water bodies include the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Water Index (NDWI) [56]. However, the modified NDWI (MNDWI) is often more effective at distinguishing water from algae or shadows [see (3)] [37], [57]. Initial land–water labels were generated using MNDWI (Bands B3 and B11), with thresholds determined through iterative testing

$$NDVI = (NIR - Red)/(NIR + Red) \quad (1)$$

$$NDWI = (Green - NIR)/(Green + NIR) \quad (2)$$

$$MNDWI = (Green - SWIR)/(Green + SWIR). \quad (3)$$

Despite its utility, MNDWI can be limited in highly turbid waters. As illustrated in Fig. 3, a tidal flat (black box) and open water (red box) exhibit identical MNDWI values (0.82), making them spectrally indistinguishable via fixed thresholds. To minimize labeling errors in these zones, manual visual interpretation and correction were performed. The final training dataset integrated Sentinel-2 bands (B2, B3, B4, B8, B11) with NDVI and MNDWI layers. This dataset was partitioned into 2592 patches (256 × 256 pixels) and split into training (2074), validation (259), and test (259) sets at an 8:1:1 ratio.

a) Modified U-NET architecture: Convolutional neural networks (CNNs), such as U-Net, SegNet, and DeepLabv3 +, are widely utilized for semantic segmentation in remote sensing [58], [59], [60], [61]. The standard U-Net employs a symmetric encoder–decoder structure with skip connections to preserve spatial details. To adapt this architecture for sea–land segmentation, we introduced two modifications (Fig. 4): a pixel self-attention (PSA) module within the skip connections

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:31:11 UTC from IEEE Xplore. Restrictions apply.

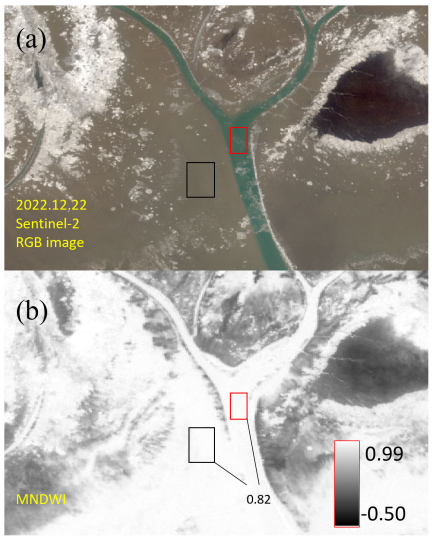


Fig. 3. (a) Sentinel-2 RGB image of the tidal flat. (b) MNDWI derived from Sentinel-2.

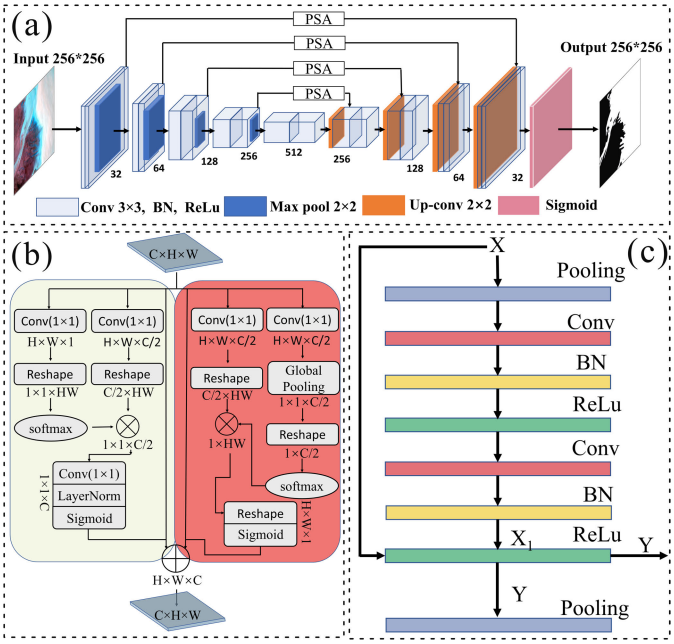


Fig. 4. (a) Modified U-Net architecture. (b) PSA module. (c) Short connection details.

and a short connection (plus layer) between adjacent modules. These additions aim to refine feature mapping and maintain performance as network depth increases [62], [63].

The PSA module [Fig. 4(b)] integrates channel and spatial self-attention to leverage interfeature relationships. For an input X, the channel self-attention weight W_C is defined as follows:

$$W_C = F_{SG}\left[F_{C1}\left(F_{R1}\left(F_{C2}\left(X\right)\right) \otimes F_{SM}\left(F_{R2}\left(F_{C3}\left(X\right)\right)\right)\right)\right] \quad (4)$$

where $F_{C1\text{-}3}$ denote 1×1 convolutions, $F_{R1\text{-}2}$ are reshape operators, $\otimes$ is the matrix dot product, and F_{SG} and F_{SM} represent the Sigmoid and Softmax operators, respectively.

The spatial self-attention weight W_S is expressed as follows:

$$W_S = F_{SG}\left[F_{R1}\left(F_{R2}\left(F_{C1}\left(X\right)\right) \otimes F_{SM}\left(F_{R3}\left(F_{GP}\left(F_{C2}\left(X\right)\right)\right)\right)\right)\right] \quad (5)$$

where F_{GP} is the global pooling operator and $F_{R1\text{-}3}$ are reshape operators.

The total output of the PSA module is the following equation:

$$Z_{PSA} = X \odot^{ch} W_C + X \odot^{SP} W_S \quad (6)$$

where $\odot^{ch}$ and $\odot^{SP}$ denote channelwise and spatialwise multiplication, and + represents elementwise addition.

The plus layer short connection, indicated by the red-dotted box in Fig. 4(a), is implemented within downsampling and upsampling modules. For an input X and output Y, the operation is the following equation:

$$Y = F_{C2}\left(\sigma\left(F_{C1}\left(\sigma\left(X\right)\right)\right)\right) + X \quad (7)$$

where $F_{C1\text{-}2}$ are convolution operators, and σ is the ReLu function. In the encoder, the output Y is passed to the subsequent module and simultaneously routed to the corresponding decoder layer via the PSA module.

b) Model training: The DL model was implemented using the PyTorch framework. Training utilized binary cross-entropy (BCE) as the loss function with an initial learning rate of 1e-4 [64]. To prevent overfitting, an early stopping mechanism was employed, halting training if the validation loss failed to decrease for 20 consecutive epochs within a maximum of 500 epochs. The input data integrated Sentinel-2 bands (B2, B3, B4, B8, B11) with MNDWI and NDVI layers.

Model performance was evaluated using Precision, Recall, and $F1$-score

$$Precision = \frac{TP}{TP + FP} \quad (8)$$

$$Recall = \frac{TP}{TP + FN} \quad (9)$$

$$F1 - score = \frac{2 \times Precison \times Recall}{Precison + Recall} \quad (10)$$

where TP, FP, and FN represent true positives, false positives, and false negatives, respectively.

Following training, the Sentinel-2 imagery was processed for sea–land segmentation. Consistent with the training phase, images were partitioned into 256 × 256 pixel patches for inference and subsequently reassembled to maintain original georeferencing. Finally, an inundation frequency map was generated by stacking the classified time-series images. For each pixel, the inundation frequency was calculated as the ratio of water-classified observations to the total number of valid observations.

2) Prior Elevation Extraction: Between 2022 and 2023, 16 ICESat-2 tracks intersected the study area, five of which were retained after screening (Fig. 1). Elevation data are referenced to the WGS84 ellipsoid. ATL03 photons were matched with inundation frequency pixels based on geographic coordinates. Only photons with high signal confidence (levels 3 or 4) were selected [43].

Outliers were removed using a three-stage filtering process: (1) the 3-sigma criterion; (2) median filtering to eliminate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:31:11 UTC from IEEE Xplore. Restrictions apply.

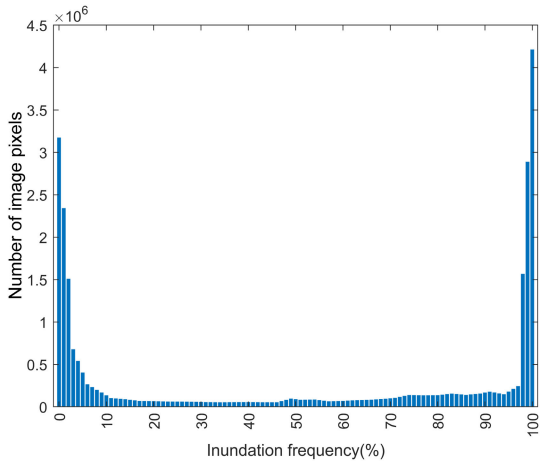


Fig. 5. Histogram of Sentinel-2 pixels by inundation frequency.

localized noise [65]; and (3) a slope-based filter to remove points with abnormal elevation gradients inconsistent with tidal flat morphology [66]. For pixels containing multiple photons, the average elevation was calculated. To maintain data integrity, ICESat-2 data over nontidal features, such as salt pans and residential structures, were manually excluded to avoid elevation errors in the final model.

3) Inversion Model Construction: The elevation inversion is based on the functional relationship where lower elevations correlate with higher inundation frequencies. For each pixel, the relationship between these two variables was modeled. Where multiple pixels shared the same inundation frequency, their elevation values were averaged. A moving average was then applied to ensure a monotonic decreasing trend in the elevation–frequency curve.

Field investigations and previous studies indicate that elevations in the study area are generally below 1.5 m, with distinct longitudinal terrain variations. To account for these spatial differences, a localized zonal modeling approach was implemented. The study area was partitioned into zones using major tidal channels and ICESat-2 orbital positions as boundaries. To ensure spatial continuity, elevation data were averaged across adjacent zones before constructing the final inversion model for each area.

III. Results

A. Inundation Frequency

Tidal flat delineation using inundation frequency typically requires defining upper and lower thresholds, often determined by the specific characteristics of the study area. Analysis of the Sentinel-2 time series revealed a high concentration of pixels with frequencies below 3% and above 97% (Fig. 5). These intervals were identified as permanent water and land areas above the high-tide line, respectively. Consequently, the 3%–97% frequency range was defined as the tidal flat zone.

The trained DL model was applied to all Sentinel-2 imagery from 2022 to 2023 to generate the inundation frequency map, with man-made structures manually masked (Fig. 6). For each pixel, the frequency was calculated as the ratio of water-classified observations to the total number of valid

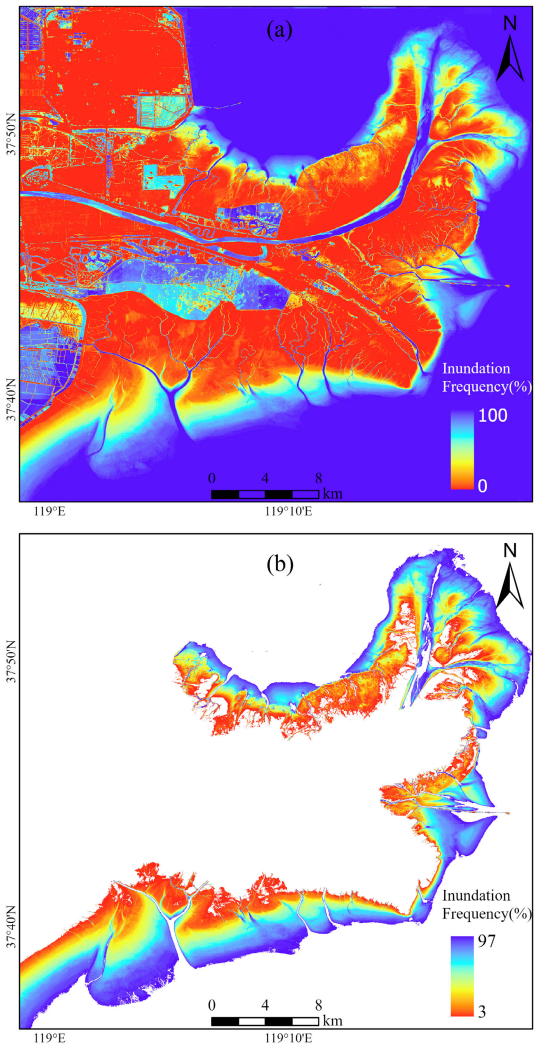


Fig. 6. (a) Inundation frequency map. (b) Tidal flat map.

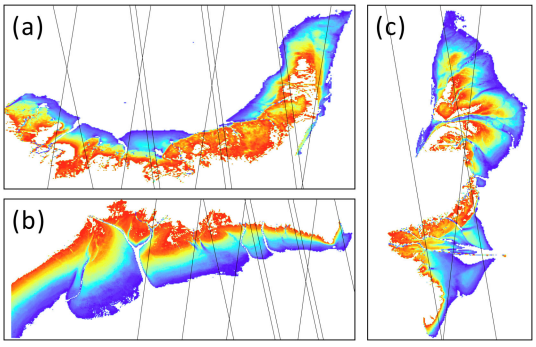


Fig. 7. Study area zonation (a) northern, (b) southern, and (c) eastern regions. Solid black lines denote ICESat-2 tracks.

observations. As illustrated in Fig. 7, the study area was partitioned into northern, eastern, and southern regions using the historical and contemporary river channels as boundaries [28]. Regional inversion models were then independently constructed and applied to each zone.

B. Elevation Mapping

Once the relationship between inundation frequency and height was established, a cubic polynomial fitting method

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:31:11 UTC from IEEE Xplore. Restrictions apply.

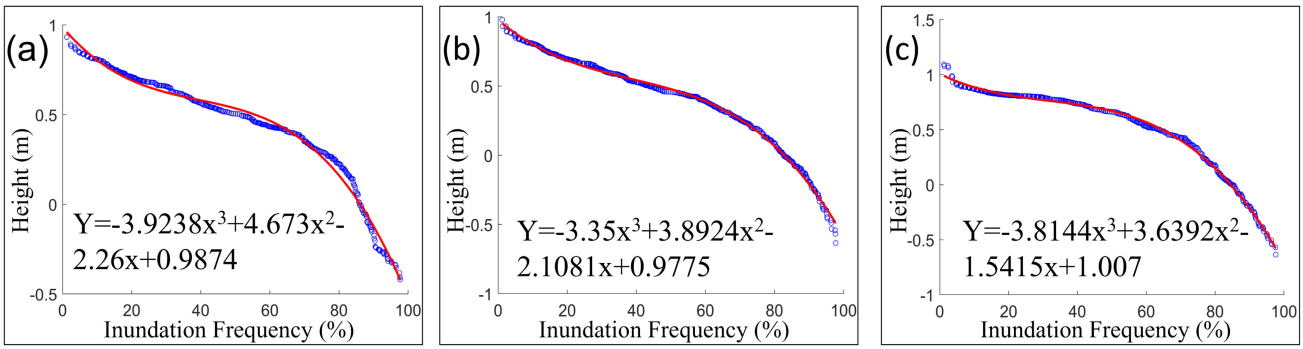


Fig. 8. (a)–(c) Relationship between inundation frequency and elevation across the northern, eastern, and southern subregions. Blue circles denote ICESat-2 observations; red lines indicate the derived functional models.

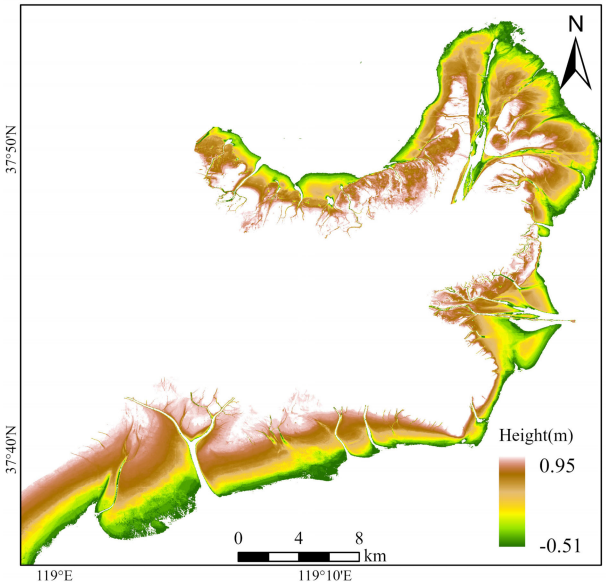


Fig. 9. Tidal flat DEM of the Yellow River estuary.

was employed to construct the inversion model using the inundation frequency and prior elevations (Fig. 8). The results indicate that the cubic polynomial fitting represents the observed data across the frequency range. After filtering, the tidal flat within the study area covers an extent of 213.68 km^2.

As shown in Fig. 9, the final DEM with a spatial resolution of 10 m was derived. The elevations across the entire study area range from −0.51 to 0.96 m. The tidal flat areas are more extensive in the southern and northeastern parts of the study area. In the southern region, the tidal flats are distributed with an east-west alignment and show a gradual elevation increase from south to north. In contrast, tidal flats in other areas do not exhibit this specific shore-parallel zonation. The tidal flats in the eastern and northern regions are smaller in extent, with relatively higher elevations and more pronounced topographical fluctuations observed in the eastern region.

C. Performance Evaluation

We analyzed the impact of different data inputs on model performance. Table II lists the input combinations, the training epochs at which early stopping occurred, and the corresponding evaluation metrics. The results indicate that using only the original Sentinel-2 bands (B2, B3, B4, B8) yields lower performance metrics compared with combinations including spectral indices. The integration of the water index (MNDWI) or vegetation index (NDVI) resulted in higher Precision and Recall values, with the MNDWI-enhanced input achieving the highest scores among the tested combinations.

TABLE II
ACCURACY COMPARISONS FOR DIFFERENT INPUTS

Input	Epoch	Accuracy metrics (%)		
		Precision	Recall	F1-score
B2, B3, B4	379	92.5	92.3	92.4
B3, B4, B8	424	93.2	92.6	92.9
B2, B3, B4, B8	315	92.4	92.8	92.6
B2, B3, B4, B8, MNDWI, NDVI	469	94.1	93.9	94.0
B2, B3, B4, B8, NDVI	327	95.7	94.4	95.0
B2, B3, B4, B8, MNDWI	412	**96.4**	**95.8**	**96.1**

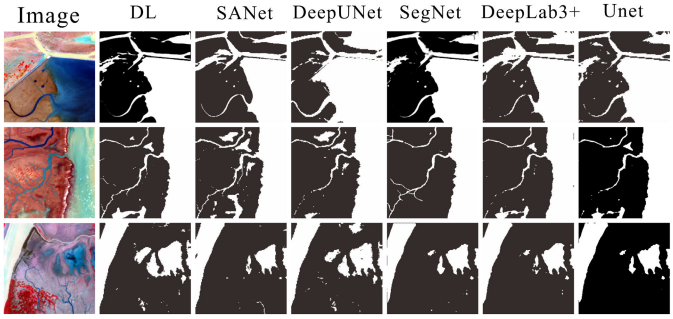


Fig. 10. Water–land segmentation results across different DL models.

A comparative analysis was also conducted across different DL architectures, including U-Net, SegNet, DeepLabv3+, DeepUNet, SANet, and the modified U-Net proposed in this study. The sea–land segmentation results for these models on the test dataset are illustrated in Fig. 10. All models were evaluated using identical datasets and hyperparameters. As

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:31:11 UTC from IEEE Xplore. Restrictions apply.

TABLE III
EVALUATION RESULTS OF DIFFERENT DL MODELS

DL models	Accuracy metrics (%)		
	Precision	Recall	F1-score
U-Net	92.6	93.2	92.9
SegNet	94.1	93.8	93.9
DeepLabv3+	93.4	93.8	93.6
DeepUNet	95.6	94.3	94.9
SANet	95.5	95.0	95.2
This study	**96.4**	**95.8**	**96.1**

presented in Table III, the modified U-Net achieved a precision of 96.4%, a recall of 95.8%, and an $F1$-score of 96.1%.

We evaluated the vertical performance of the satellite-derived tidal flat DEMs using two independent validation datasets: UAV LiDAR and ICESat-2 altimetry. To ensure geometric consistency, the vertical reference for all datasets was unified to the WGS84 ellipsoid, allowing for an assessment based on absolute elevation differences. Nonparametric robust estimation metrics were employed to evaluate elevation residuals. These metrics provide resistance to outliers and adaptability to asymmetric data distributions, offering a characterization of error distributions [21], [67], [68]. Raw UAV LiDAR point clouds were preprocessed to establish a terrain reference. Ground points were isolated using the progressive triangular irregular network densification (PTD) algorithm to exclude vegetation and anthropogenic structures, then interpolated into a 0.5-m resolution DEM via inverse distance weighting (IDW). For comparison with the 10-m Sentinel-2-derived DEMs, the LiDAR-based elevation for each 10-m pixel was calculated as the arithmetic mean of its constituent 0.5-m cells.

Validation against 14 887 LiDAR-derived pixels yielded an RMSE of 0.18 m and a Spearman correlation of 0.78 (Fig. 11). The mean and median elevation differences were both 0.04 m, with a median absolute deviation (MAD) of 0.12 m, a normalized median absolute deviation (NMAD) of 0.20 m, and a linear error at the 90% confidence level (LE90) of 0.28 m. Comparison with 16 879 ICESat-2 pixels resulted in an RMSE of 0.18 m and a Spearman correlation of 0.80. The mean difference was 0.04 m, the median was 0.03 m, the MAD was 0.13 m, the NMAD was 0.19 m, and the LE90 was 0.29 m. These results indicate that the method captures delta-scale tidal flat topography at a 10-m resolution. Error distributions were approximately normal and symmetric, with most residuals concentrated between −0.3 and 0.3 m. The Q–Q and box plots further confirm that while outliers exist, the elevation residuals generally follow a normal distribution.

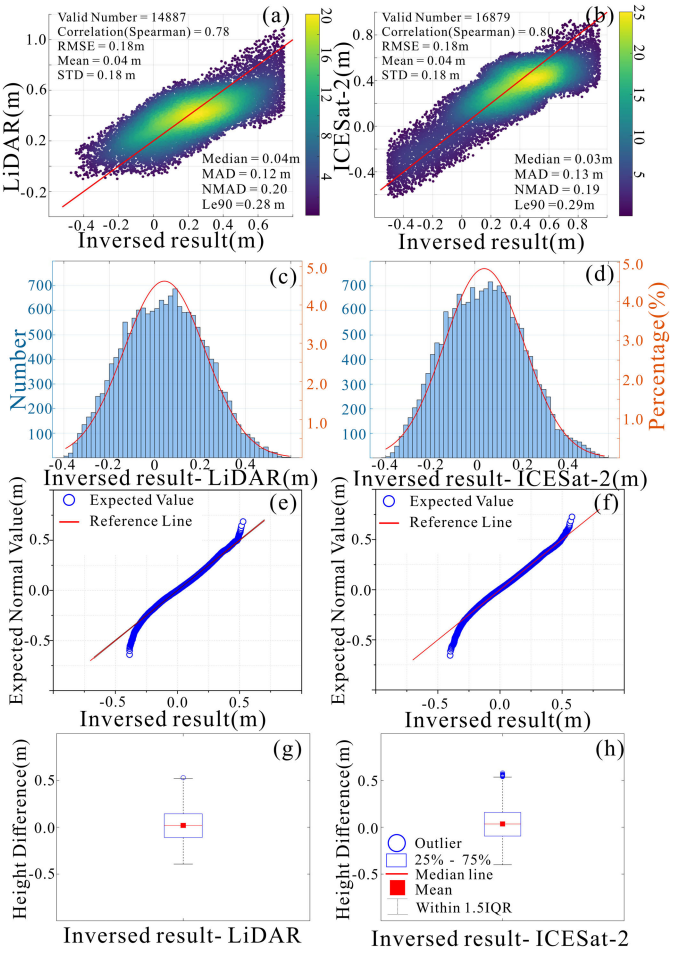


Fig. 11. Elevation validation against UAV LiDAR and ICESat-2. (a) and (b) Correlation scatter plots. (c) and (d) Error histograms. (e) and (f) Q–Q plots. (g) and (h) Boxplots.

IV. DISCUSSION

A. Impact of Estuarine Characteristics

Figs. 7 and 9 characterize the geomorphological features of the Yellow River estuary, where interwoven tidal flats and channels create a complex land–water interface. Low-lying areas exhibit higher inundation frequencies, while elevated regions display lower frequencies, reflecting the interplay of tidal forces, sediment deposition, and river erosion. High-frequency areas are primarily situated near channel mouths and tributaries, identifying the pathways of tidal flow and sediment transport.

The UAV LiDAR validation in the eastern estuary yielded performance metrics consistent with the ICESat-2 validation across the study area. This indicates that despite the hydrodynamic complexity of northern and eastern regions, the method maintains vertical performance comparable to the stable southern tidal flats. However, localized discrepancies occur near complex coastlines and narrow channels, where the 10-m spatial resolution limit leads to mixed-pixel effects. These effects introduce uncertainties in water extraction, resulting in minor elevation residuals. Additionally, as the derived DEM represents the average topographic state over the two-year observation period, it does not capture intraperiod morphological changes driven by sediment dynamics. The deviations between this time-averaged representation and instantaneous elevations in validation data contribute to the observed elevation residuals in dynamic areas.

B. Effectiveness of DL

The identification of tidal flats remains challenging due to fluctuating water levels, vegetation cover, and reflectance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:31:11 UTC from IEEE Xplore. Restrictions apply.

variability driven by tidal cycles. These dynamic characteristics reduce the spectral contrast between water and land, often limiting the performance of traditional methods based solely on spectral indices in complex environments such as the Yellow River estuary [38]. To address these constraints, we utilized a modified U-Net model to perform pixel-level waterline segmentation and capture spatial details.

Two modifications were introduced to the U-Net architecture to enhance segmentation performance. First, a PSA module was integrated into the skip connections to enable dynamic pixelwise context capture for context-aware segmentation. Second, a plus layer was added as a short connection between adjacent downsampling and upsampling modules to improve information flow and maintain feature continuity as network depth increases.

Comparisons with the original U-Net architecture show an increase in the $F1$-score from 92.9% to 96.1% (Table III). The modified network also yielded higher performance metrics than several baseline models, particularly in intertidal zones with dynamic land–water boundaries. These findings indicate that integrating attention mechanisms and structural enhancements can improve segmentation results in complex remote sensing applications.

C. Spatial Heterogeneity and Inundation Frequency

Initial model construction considered a single inversion for the entire study area. However, ICESat-2 observations revealed spatial heterogeneity, where elevations corresponding to the same inundation frequency differed between the southern and northern coasts due to tidal variability. This divergence indicated that the inundation frequency–elevation relationship is not spatially invariant, suggesting that a single global model could introduce regional biases. Consequently, we adopted a zonal modeling approach. Zonation was applied where reference data indicated that elevation–frequency relationships varied across regions, leading to the division of the study area into three subregions. ICESat-2 observations from each zone were averaged with adjacent data to ensure transitions, and the resulting values were used for modeling. To evaluate this strategy, a unified model was constructed by aggregating observations across all regions. Validation against airborne LiDAR showed that the global model yielded an RMSE of 0.28 m, while the zonal approach achieved an RMSE of 0.18 m. This 0.10 m reduction indicates that zonal modeling mitigates residuals caused by spatial tidal variability.

The representation of inundation frequency from remote sensing depends on temporal resolution and the range of tidal conditions captured. While the Sentinel-2 five-day revisit cycle supports time-series monitoring, fixed acquisition times and cloud cover can limit the representation of peak tidal dynamics. Integrating additional data sources, such as Sentinel-1 SAR and Landsat imagery, may address these constraints [39], [40]. Sentinel-1 provides all-weather imaging, while the Landsat archive enhances temporal coverage, potentially enabling a more detailed depiction of tidal variations.

The 2022–2023 inundation frequency map and derived DEM were constructed assuming topographic stability over the two-year period. However, episodic events, such as the October 2022 storm surge, likely altered the morphology, introducing elevation residuals. While longer data spans increase the likelihood of such temporal decorrelation, shorter periods are limited by image availability. Furthermore, cloud cover can hinder comprehensive observation, affecting the temporal resolution of the resulting DEM. Future research will focus on integrating multisource data to address these constraints and enable higher frequency monitoring.

D. Vertical Reference and Data Quality

The generation of a tidal flat DEM at the delta scale depends on the selection and processing of prior elevation data [33], [39], [45]. To maintain the reliability of the ICESat-2 dataset, several technical measures were implemented.

First, the tidal stage was the primary selection criterion. Due to the high turbidity of the Yellow River estuary, the laser penetration of ICESat-2 is limited. Under high-tide conditions, returned photons primarily represent the water surface as bottom returns are significantly attenuated. To mitigate this effect, ICESat-2 tracks were strictly filtered to include only those acquired during low-tide windows, ensuring measurements correspond to the exposed tidal flat surface. Second, the spatial coverage and SNR of the selected low-tide tracks were evaluated. Tracks with insufficient sampling density or a high proportion of low-confidence photon flags were excluded to maintain data consistency. Finally, to address residual noise such as solar background photons and transient outliers, we applied median and slope-based filtering. These algorithms identified and removed residual elevation anomalies, thereby improving the accuracy of the final elevation reference.

E. Methodological Transferability

To assess the generalizability of the proposed method, it was applied to the radial sand ridge system in the South Yellow Sea, a macroscale geomorphic feature covering 22 000 km^2 (Fig. 12). Dynamically linked to the abandoned Yellow River Delta as a primary sediment source [69], this intertidal system consists of over 70 radiating ridges and channels shaped by converging tidal waves [70]. Unlike the Yellow River estuary, this region represents a complex mudflat system with distinct sedimentation-erosion dynamics. Without parameter adjustment, the method yielded an RMSE of 0.52 m. While this vertical performance may limit microtopographic detection, it demonstrates the transferability of the workflow for first-order elevation representation in diverse coastal environments.

As shown in Fig. 12, the error distribution is near-normal with a median of −0.39 m and an interquartile range from −0.75 to −0.1 m. The Q–Q plot indicates that the central errors align with a theoretical normal distribution, though heavier tails are present. The observed slight underestimation of elevations may be related to seasonal vegetation effects. The ICESat-2 data used for model construction were potentially acquired during periods of senescent vegetation, whereas validation data may have captured active growth phases where laser returns reflect from the canopy rather than the bare surface. This discrepancy could introduce a systematic negative bias in the error distribution, although this remains a hypothesis requiring further investigation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:31:11 UTC from IEEE Xplore. Restrictions apply.

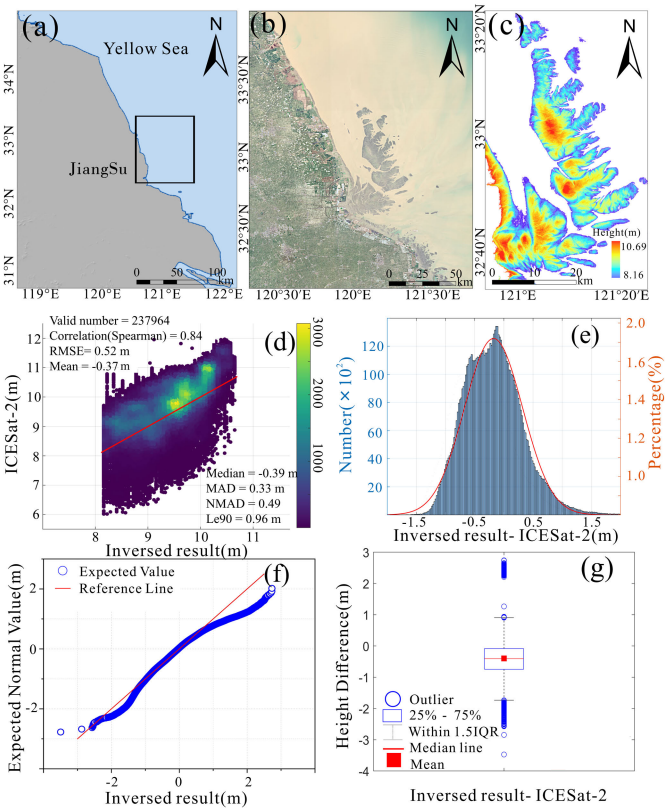


Fig. 12. Tidal flat DEM and accuracy evaluation in the radial sand ridge system in the South Yellow Sea. (a) Geographic location. (b) Optical satellite image from Sentinel-2. (c) Final DEM map. (d) Correlation scatter plot. (e) Histogram. (f) Q-Q plot. (e) Box plot.

This transfer process required no region-specific tuning, indicating a capacity for zero-parameter application across different tidal regimes. By integrating Sentinel-2 optical imagery with ICESat-2 altimetry, this method establishes a scalable framework for generating intertidal DEMs. Future research will evaluate performance across more diverse contexts such as lagoons and mangroves. Incorporating higher resolution imagery and in situ measurements could also enhance the capacity to capture short-term morphological dynamics and account for vegetation-related disturbances.

V. Conclusion

In this study, we developed a method to reconstruct delta-scale tidal flat topography in highly turbid estuaries by integrating satellite remote sensing and DL. By addressing the limitations of traditional survey methods in dynamic and turbid environments, this approach enables systematic topographic reconstruction at a 10-m spatial resolution. The resulting elevation models characterize tidal flat morphology, providing a data-driven basis for analyzing coastal processes and ecological habitats. By leveraging the revisit frequency of Sentinel-2 alongside ICESat-2 observations, this method operates independently of field measurements, making it applicable to other coastal regions with similar satellite data availability. Validation in both the Yellow River Delta and the radial sand ridge system in the South Yellow Sea demonstrated consistent performance, suggesting that the model maintains its predictive capacity across different geographic settings. Overall, this approach provides a scalable framework for periodic monitoring of intertidal environments at the delta scale.

The resulting DEM captures macrotopographic patterns and tidal zone dynamics, supporting the integration of topographic data into ecological and anthropogenic studies such as erosion risk assessment and reclamation impacts. While the current method demonstrates robustness using publicly available datasets, future research will aim to further refine the detection of subpixel geomorphological changes by exploring the integration of submeter commercial imagery. Furthermore, as high-resolution SAR missions, such as ICEYE and Capella Space, become more accessible, investigating multisource data fusion by combining the structural insights of SAR with the spectral detail of optical sensors could overcome current resolution-related constraints and provide monitoring capabilities across a wider range of global geomorphic and climatic conditions.

Acknowledgment

The authors would like to thank the anonymous reviewers and the Editor for their constructive comments, which improved the quality of this article. They also thank colleagues in the Coastal Remote Sensing Group at Ocean University of China for their assistance with field surveys and data analysis. Sentinel-2 and ICESat-2 datasets are publicly available from USGS and NASA.

References

[1] N. J. Murray et al., "The global distribution and trajectory of tidal flats," *Nature*, vol. 565, no. 7738, pp. 222–225, Jan. 2019.
[2] S. Liu et al., "Dynamics and drivers of tidal flat morphology in China," *Nature Commun.*, vol. 16, no. 1, p. 2153, Mar. 2025.
[3] Z. Ma et al., "Rethinking China's new great wall," *Science*, vol. 346, no. 6212, pp. 912–914, Nov. 2014.
[4] S. Ghosh, D. R. Mishra, and A. A. Gitelson, "Long-term monitoring of biophysical characteristics of tidal wetlands in the northern Gulf of Mexico—A methodological approach using MODIS," *Remote Sens. Environ.*, vol. 173, pp. 39–58, Feb. 2016.
[5] H. K. Lotze et al., "Depletion, degradation, and recovery potential of estuaries and coastal seas," *Science*, vol. 312, no. 5781, pp. 1806–1809, Jun. 2006.
[6] Y. Wang, P. Li, N. Bi, X. Wu, and H. Wang, "Distribution pattern of carbon storage by native vegetation in the Yellow River estuary," *Estuarine, Coastal Shelf Sci.*, vol. 332, May 2026, Art. no. 109740.
[7] S. Temmerman, P. Meire, T. J. Bouma, P. M. J. Herman, T. Ysebaert, and H. J. De Vriend, "Ecosystem-based coastal defence in the face of global change," *Nature*, vol. 504, no. 7478, pp. 79–83, Dec. 2013.
[8] M. L. Kirwan and J. P. Megonigal, "Tidal wetland stability in the face of human impacts and sea-level rise," *Nature*, vol. 504, no. 7478, pp. 53–60, Dec. 2013.
[9] M. Schuerch et al., "Future response of global coastal wetlands to sea-level rise," *Nature*, vol. 561, no. 7722, pp. 231–234, Sep. 2018.
[10] J. Liu et al., "Spatiotemporal change detection of coastal wetlands using multi-band SAR coherence and synergetic classification," *Remote Sens.*, vol. 14, no. 11, p. 2610, May 2022.
[11] P. Li, G. Wang, C. Liang, H. Wang, and Z. Li, "InSAR-derived coastal subsidence reveals new inundation scenarios over the Yellow River Delta," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 8431–8441, 2023.
[12] S. L. Yang, K. H. Xu, J. D. Milliman, H. F. Yang, and C. S. Wu, "Decline of Yangtze River water and sediment discharge: Impact from natural and anthropogenic changes," *Sci. Rep.*, vol. 5, no. 1, p. 12581, Jul. 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:31:11 UTC from IEEE Xplore. Restrictions apply.

[13] J. Syvitski et al., "Large deltas, small deltas: Toward a more rigorous understanding of coastal marine deltas," *Global Planet. Change*, vol. 218, Nov. 2022, Art. no. 103958.
[14] J. H. Nienhuis et al., "Global-scale human impact on delta morphology has led to net land area gain," *Nature*, vol. 577, no. 7791, pp. 514–518, Jan. 2020.
[15] J. P. M. Syvitski et al., "Sinking deltas due to human activities," *Nature Geosci.*, vol. 2, no. 10, pp. 681–686, Oct. 2009.
[16] L. Giosan, J. Syvitski, S. Constantinescu, and J. Day, "Climate change: Protect the world's deltas," *Nature*, vol. 516, no. 7529, pp. 31–33, Dec. 2014.
[17] G. Wang et al., "Coastal dam inundation assessment for the Yellow River delta: Measurements, analysis and scenario," *Remote Sens.*, vol. 12, no. 21, p. 3658, Nov. 2020.
[18] G. Wang, P. Li, Z. Li, C. Liang, and H. Wang, "Coastal subsidence detection and characterization caused by brine mining over the Yellow River delta using time series InSAR and PCA," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 114, Nov. 2022, Art. no. 103077.
[19] G. Wang, P. Li, Z. Li, J. Liu, Y. Zhang, and H. Wang, "InSAR and machine learning reveal new understanding of coastal subsidence risk in the Yellow River delta, China," *Sci. Total Environ.*, vol. 915, Mar. 2024, Art. no. 170203.
[20] R. J. Nicholls and M. Shirzaei, "Earth's sinking surface," *Science*, vol. 384, no. 6693, pp. 268–269, 2024.
[21] P. Li, Z. Li, K. Dai, Y. Al-Husseinawi, W. Feng, and H. Wang, "Reconstruction and evaluation of DEMs from bistatic tandem-X SAR in mountainous and coastal areas of China," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 5152–5170, 2021.
[22] K. Seeger and P. S. J. Minderhoud, "Sea level much higher than assumed in most coastal hazard assessments," *Nature*, vol. 652, no. 8110, pp. 667–674, Apr. 2026.
[23] L. O. Ohenhen et al., "Global subsidence of river deltas," *Nature*, vol. 649, no. 8098, pp. 894–901, Jan. 2026.
[24] J.-M. Lee, J.-Y. Park, and J.-Y. Choi, "Evaluation of sub-aerial topographic surveying techniques using total station and RTK-GPS for applications in macrotidal sand beach environment," *J. Coastal Res.*, vol. 65, pp. 535–540, Jan. 2013.
[25] C. Tu et al., "Synergetic classification of coastal wetlands over the Yellow River delta with GF-3 full-polarization SAR and zhuhai-1 OHS hyperspectral remote sensing," *Remote Sens.*, vol. 13, no. 21, p. 4444, Nov. 2021.
[26] Q. Zhu et al., "Spatiotemporal changes of coastline over the Yellow River delta in the previous 40 years with optical and SAR remote sensing," *Remote Sens.*, vol. 13, no. 10, p. 1940, May 2021.
[27] M. Chang, P. Li, Z. Li, and H. Wang, "Mapping tidal flats of the Bohai and Yellow Seas using time series Sentinel-2 images and Google Earth engine," *Remote Sens.*, vol. 14, no. 8, p. 1789, Apr. 2022.
[28] P. Li, S. Li, Z. Li, C. Liang, and H. Wang, "Detailed detection and extraction of estuarine tidal channels with multispectral and full-polarised SAR remote sensing," *Earth Surf. Processes Landforms*, vol. 49, no. 12, pp. 3968–3988, Sep. 2024.
[29] P. Li, Y. Zhang, C. Liang, H. Wang, and Z. Li, "High spatiotemporal resolution river networks mapping on catchment scale using satellite remote sensing imagery and DEM data," *Geophys. Res. Lett.*, vol. 51, no. 6, pp. e2023GL10795–6, Mar. 2024.
[30] E. Salameh, F. Frappart, I. Turki, and B. Laignel, "Intertidal topography mapping using the waterline method from Sentinel-1 & -2 images: The examples of arcachon and veys bays in France," *ISPRS J. Photogramm. Remote Sens.*, vol. 163, pp. 98–120, May 2020.
[31] E. Salameh et al., "Evaluating SWOT's interferometric capabilities for mapping intertidal topography," *Remote Sens. Environ.*, vol. 314, Dec. 2024, Art. no. 114401.
[32] H. Xu et al., "A large-scale framework for deriving tidal flat topography from SWOT data," *Remote Sens. Environ.*, vol. 334, Apr. 2026, Art. no. 115237.
[33] C. Chen, C. Zhang, B. Tian, W. Wu, and Y. Zhou, "Tide2Topo: A new method for mapping intertidal topography accurately in complex estuaries and bays with time-series Sentinel-2 images," *ISPRS J. Photogramm. Remote Sens.*, vol. 200, pp. 55–72, Jun. 2023.
[34] D. C. Mason, I. J. Davenport, G. J. Robinson, R. A. Flather, and B. S. McCartney, "Construction of an inter-tidal digital elevation model by the 'water-line' method," *Geophys. Res. Lett.*, vol. 22, no. 23, pp. 3187–3190, Dec. 1995.
[35] Y. Kang et al., "Topographic mapping on large-scale tidal flats with an iterative approach on the waterline method," *Estuarine, Coastal Shelf Sci.*, vol. 190, pp. 11–22, 2017/05/05/, 2017.
[36] J. de Vries, B. van Maanen, G. Ruessink, P. A. Verweij, and S. M. de Jong, "Unmixing water and mud: Characterizing diffuse boundaries of subtidal mud banks from individual satellite observations," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 95, Mar. 2021, Art. no. 102252.
[37] C. Chen, C. Zhang, B. Tian, W. Wu, and Y. Zhou, "Mapping intertidal topographic changes in a highly turbid estuary using dense Sentinel-2 time series with deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 205, pp. 1–16, Nov. 2023.
[38] S. Zhang, Q. Xu, H. Wang, Y. Kang, and X. Li, "Automatic waterline extraction and topographic mapping of tidal flats from SAR images based on deep learning," *Geophys. Res. Lett.*, vol. 49, no. 2, Jan. 2022, Art. no. e2021GL096007.
[39] S. Yao, K. Tan, Y. Wang, W. Zhang, S. Liu, and J. Yang, "Estimating terrain elevations at 10 m resolution by integrating random forest machine learning model and ICESat-2, Sentinel-1, and Sentinel-2 satellite remotely sensed data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 132, Aug. 2024, Art. no. 104010.
[40] P. Tang et al., "Tidal flat topography mapping with Sentinel time series using cross-modal sample transfer and deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 225, pp. 69–87, Jul. 2025.
[41] Z. Niu, J. Wang, M. Wu, and Q. Han, "Remote-sensing monitoring and analysis of China intertidal zone changes based on tidal correction," *Chin. Sci. Bull.*, vol. 64, no. 4, pp. 456–473, Dec. 2018.
[42] Y. Liu, M. Li, M. Zhou, K. Yang, and L. Mao, "Quantitative analysis of the waterline method for topographical mapping of tidal flats: A case study in the Dongsha Sandbank, China," *Remote Sens.*, vol. 5, no. 11, pp. 6138–6158, Nov. 2013.
[43] T. A. Neumann et al., "The ice, cloud, and land elevation satellite-2 mission: A global geolocated photon product derived from the advanced topographic laser altimeter system," *Remote Sens. Environ.*, vol. 233, Nov. 2019, Art. no. 111325.
[44] Y. Ma et al., "Satellite-derived bathymetry using the ICESat-2 LiDAR and Sentinel-2 imagery datasets," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112047.
[45] N. Xu, Y. Ma, J. Yang, X. H. Wang, Y. Wang, and R. Xu, "Deriving tidal flat topography using ICESat-2 laser altimetry and Sentinel-2 imagery," *Geophys. Res. Lett.*, vol. 49, no. 2, pp. e2021GL09681–3, Jan. 2022.
[46] K. Wang, H. Li, N. Zhang, J. Zhang, X. Zhang, and Z. Gong, "Study on the erosion and deposition changes of tidal flat in Jiangsu province using ICESat-2 and Sentinel-2 data," *Remote Sens.*, vol. 15, no. 14, p. 3598, Jul. 2023.
[47] X. Zheng, B. Zhou, H. Lei, Y. Jin, and Q. Su, "Mapping tidal flat topography using time-series Sentinel-2 images and ICESat-2 data: A case study in cixi city," *Ocean Coastal Manage.*, vol. 256, Oct. 2024, Art. no. 107278.
[48] J.-F. Pekel, A. Cottam, N. Gorelick, and A. S. Belward, "High-resolution mapping of global surface water and its long-term changes," *Nature*, vol. 540, no. 7633, pp. 418–422, Dec. 2016.
[49] H. Xin et al., "Mapping tidal flat topography by combining ICESat-2 laser altimetry and multi-source satellite imagery," *Int. J. Digit. Earth*, vol. 18, no. 2, Dec. 2025, Art. no. 2554313.
[50] M. Chang et al., "Time series satellite remote sensing reveals annual distribution, composition, and trajectory of tidal wetlands in the Yellow River delta," *Estuarine, Coastal Shelf Sci.*, vol. 319, Jul. 2025, Art. no. 109264.
[51] M. Chang, P. Li, Z. Li, and H. Wang, "An accurate algorithm for 10m monthly mapping of tidal wetlands in the Yellow River delta with satellite remote sensing," *Remote Sens. Appl. Soc. Environ.*, vol. 41, Jan. 2026, Art. no. 101921.
[52] M. Chang, P. Li, Y. Sun, H. Wang, and Z. Li, "Mapping dynamic turbidity maximum zone of the Yellow River estuary from 38 years of Landsat imagery," *Remote Sens.*, vol. 14, no. 15, p. 3782, Aug. 2022.
[53] S. Foga et al., "Cloud detection algorithm comparison and validation for operational Landsat data products," *Remote Sens. Environ.*, vol. 194, pp. 379–390, Jun. 2017.
[54] T. Neumann et al., "ATLAS/ICESat-2 L2A global geolocated photon data, version 3," NASA Nat. Snow Ice Data Center Distrib. Act. Arch. Center, Boulder, CO, USA, Tech. Rep. ATL03, 2025. Accessed: Apr. 22, 2026.
[55] B. Huang, P. Li, H. Lu, J. Yin, Z. Li, and H. Wang, "WaterDetectionNet: A new deep learning method for flood mapping with SAR image convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 14471–14485, 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:31:11 UTC from IEEE Xplore. Restrictions apply.

[56] H. Xu, "Modification of normalised difference water index (NDWI) to enhance open water features in remotely sensed imagery," *Int. J. Remote Sens.*, vol. 27, no. 14, pp. 3025–3033, Jul. 2006.

[57] B. Zhao, Y. Liu, L. Wang, Y. Liu, C. Sun, and S. Fagherazzi, "Stability evaluation of tidal flats based on time-series satellite images: A case study of the Jiangsu central coast, China," *Estuarine, Coastal Shelf Sci.*, vol. 264, Jan. 2022, Art. no. 107697.

[58] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[59] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*. Cham, Switzerland: Springer, pp. 234–241.

[60] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[61] L.-C. Chen et al., "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. ECCV*, Munich, Germany, 2018, pp. 833–851.

[62] R. Li et al., "DeepUNet: A deep fully convolutional network for pixel-level sea-land segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 3954–3962, Nov. 2018.

[63] H. Liu, F. Liu, X. Fan, and D. Huang, "Polarized self-attention: Towards high-quality pixel-wise mapping," *Neurocomputing*, vol. 506, pp. 158–167, Sep. 2022.

[64] M. Aghdami-Nia, R. Shah-Hosseini, A. Rostami, and S. Homayouni, "Automatic coastline extraction through enhanced sea-land segmentation by modifying standard U-Net," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 109, May 2022, Art. no. 102785.

[65] T. Xie, R. Kong, A. Nurunnabi, S. Bai, and X. Zhang, "Machine-learning-method-based inversion of shallow bathymetric maps using ICESat-2 ATL03 data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3697–3714, 2023.

[66] G. Vosselman, "Slope based filtering of laser altimetry data," *Proc. Int. Arch. Photogramm. Remote Sens. (IAPRS)*, vol. 33, Amsterdam, The Netherlands, 2000, pp. 935–942.

[67] J. Höhle and M. Höhle, "Accuracy assessment of digital elevation models by means of robust statistical methods," *ISPRS J. Photogramm. Remote Sens.*, vol. 64, no. 4, pp. 398–406, Jul. 2009.

[68] P. Li et al., "Large-scale high-resolution coastal subsidence mapping in eastern China with Sentinel-1 and Sentinel-2: Heterogeneous patterns and primary drivers," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 146, Feb. 2026, Art. no. 105047.

[69] Y. Wang, Y. Zhang, X. Zou, D. Zhu, and D. Piper, "The sand ridge field of the south Yellow Sea: Origin by river–sea interaction," *Mar. Geol.*, vols. 291–294, pp. 132–146, Jan. 2012.

[70] J. Tao, Z. B. Wang, Z. Zhou, F. Xu, C. Zhang, and M. J. F. Stive, "A morphodynamic modeling study on the formation of the large-scale radial sand ridges in the Southern Yellow Sea," *J. Geophys. Res., Earth Surf.*, vol. 124, no. 7, pp. 1742–1761, Jul. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:31:11 UTC from IEEE Xplore. Restrictions apply.